

# KW 11: The IS-LM Model

Chapters 5 & 6 + Lecture 4

*Blanchard - Macroeconomics, 7th Global Edition*

*Study Notes*

## LECTURE 4 CONTEXT: THE IS-LM MODEL

### Learning Objectives

- The economy in the short term
- The IS-LM model: combining goods and financial markets
- Shocks and economic policy (fiscal and monetary)
- The limitations of the IS-LM model

Literature: Blanchard, Macroeconomics, Chapters 5 & 6.

### IS Curve - Short Repetition

There is an equilibrium in the goods market if production  $Y$  equals the demand for goods  $Z$ .

$$Y = C(Y - T) + I(Y, i) + G \quad (\text{assumption: } EX = IM = 0)$$

- As the interest rate rises, income decreases in the goods market equilibrium
- The IS curve therefore has a **downward slope**

### The IS curve – Short repetition

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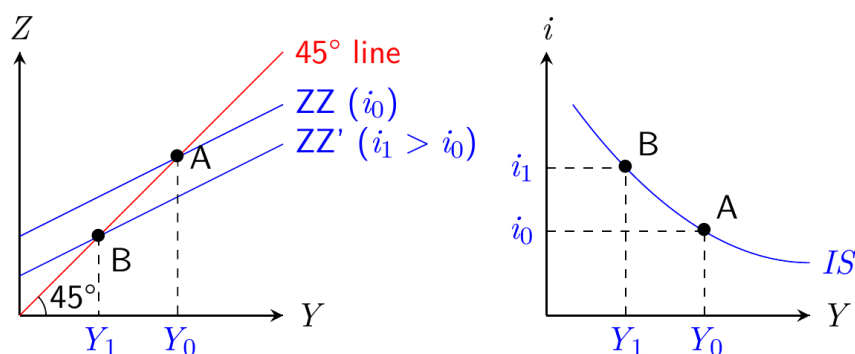


Figure: IS Curve Derivation - From goods market equilibrium (ZZ diagram) to the downward-sloping IS curve

### LM Curve - Short Repetition

The financial market is in equilibrium when the real money supply equals the real money demand:

$$\frac{M}{P} = Y \cdot L(i)$$

- We distinguish between **money supply control** (left charts) and **interest rate control** (right charts)

## The LM Curve — Short repetition

- The financial market is in equilibrium when the real money supply is equal to the real money demand:  $M/P = Y \cdot L(i)$
- We distinguish between money supply control (left-hand chart) and interest rate control (right-hand chart):

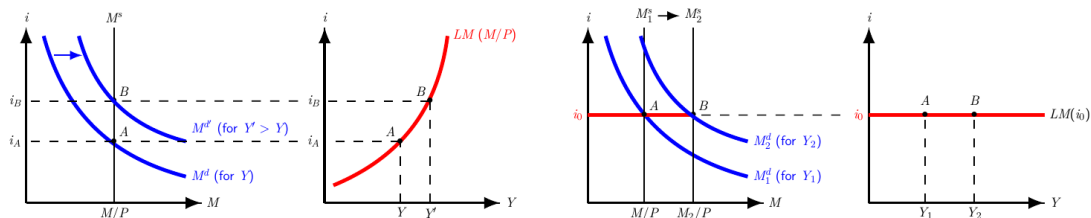


Figure: LM Curve - Money supply control (upward-sloping LM) vs. interest rate control (horizontal LM)

## IS-LM Model: Combining Both Markets

John Hicks and Alvin Hansen developed the IS-LM model in the late 1930s and early 1940s, based on Keynes' General Theory (1936).

- While in the original model the central bank controls the money supply, we usually consider the case of **interest rate control**

$$\text{IS curve: } Y = C(Y - T) + I(Y, i) + G$$

$$\text{LM curve: } i = \bar{i}$$

- Only at the intersection of both curves do we have **simultaneous equilibrium** in the goods and financial markets

## IS curve and LM curve together: IS-LM Model (Graph)

- ▶ Only in point A, the intersection of both curves, we have a simultaneous equilibrium in the goods and financial markets.
- ▶ Graphical representation of the IS-LM model:

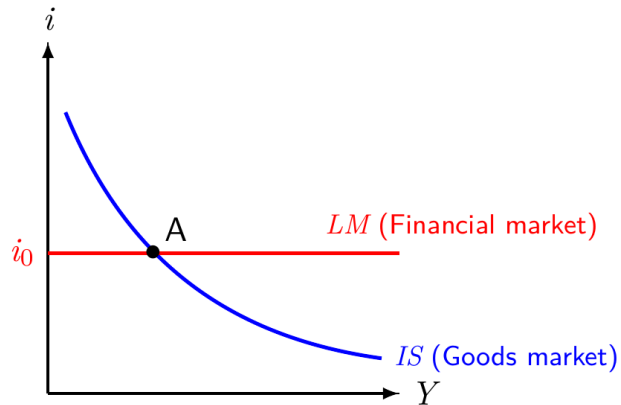


Figure: The IS-LM Model - Equilibrium at point A where both goods and financial markets clear

# CHAPTER 5: GOODS AND FINANCIAL MARKETS; THE IS-LM MODEL

## 5-1 The Goods Market and the IS Relation

In Chapter 3, we characterized equilibrium as  $Y = Z$  with constant investment. Now we relax this: investment depends on the interest rate.

### Investment, Sales, and the Interest Rate

Investment depends primarily on two factors:

- **The level of sales (Y):** A firm facing rising sales needs to invest in additional capacity
- **The interest rate (i):** Higher interest rates make borrowing more expensive, discouraging investment

$$I = I(Y, i)$$

$$(+) \quad (-)$$

Investment is an increasing function of production and a decreasing function of the interest rate.

### Determining Output

Taking investment into account, the equilibrium condition in the goods market becomes:

$$(5.2) \quad Y = C(Y - T) + I(Y, i) + G$$

This is the expanded IS relation. For a given interest rate  $i$ , demand is an increasing function of output (through both consumption and investment effects). The demand curve  $ZZ$  is upward-sloping but flatter than the 45-degree line.

### Deriving the IS Curve

- For a given interest rate, equilibrium output is where  $ZZ$  crosses the 45-degree line
- If the interest rate increases from  $i$  to  $i'$ , investment falls,  $ZZ$  shifts down to  $ZZ'$ , and equilibrium output decreases from  $Y$  to  $Y'$
- The resulting relation between  $i$  and  $Y$  is the **downward-sloping IS curve**

### Shifts of the IS Curve

- Changes in  $T$  or  $G$  shift the IS curve (they are exogenous to the IS relation)
- An increase in taxes  $T$  -> lower disposable income -> lower consumption -> IS shifts **left**
- An increase in government spending  $G$  -> higher demand -> IS shifts **right**
- An increase in consumer confidence (higher  $c_0$ ) -> IS shifts **right**

#### IS Curve Summary

The IS curve shows all combinations of  $(i, Y)$  where the goods market is in equilibrium. It is downward-sloping: higher interest rates reduce investment, leading to lower output. Changes in  $T$ ,  $G$ , or consumer confidence shift the IS curve.

## 5-2 Financial Markets and the LM Relation

From Chapter 4, the interest rate is determined by the equality of money supply and money demand:

$$M = PY \cdot L(i)$$

Rewriting in terms of real money (dividing by  $P$ ):

$$(5.3) \quad \frac{M}{P} = Y \cdot L(i)$$

This is the LM relation: the real money supply must equal real money demand.

### Deriving the LM Curve

The traditional approach: the central bank chooses the money stock  $M$ , and the interest rate adjusts. This gives an upward-sloping LM curve.

The modern approach: central banks now focus directly on the interest rate. They choose an interest rate  $\bar{i}$  and adjust the money supply to achieve it. This makes the LM curve a simple horizontal line:

$$\text{LM curve: } i = \bar{i}$$

#### LM Curve Summary

Under interest rate control (modern approach), the LM curve is a horizontal line at the policy rate  $\bar{i}$ . The central bank adjusts the money supply as needed to maintain this rate. Under money supply control (traditional), the LM curve is upward-sloping.

## 5-3 Putting the IS and LM Relations Together

$$\text{IS relation: } Y = C(Y - T) + I(Y, i) + G$$

$$\text{LM relation: } i = \bar{i}$$

- Any point on the IS curve = goods market equilibrium
- Any point on the LM curve = financial market equilibrium
- Only at their intersection (point A) are **both** markets in equilibrium simultaneously

### Three-Step Method for Policy Analysis

When analyzing effects of policy changes, always follow these three steps:

- (1) Does it shift the IS curve, the LM curve, or both?

- (2) What happens to the equilibrium (intersection of IS and LM)?
- (3) Describe the effects in words

# SHOCKS AND ECONOMIC POLICY

## Comparative Statics in the IS-LM Model

When conducting comparative statics, we investigate how equilibrium levels of  $i$  and  $Y$  adjust as we change one variable.

- We change **exogenous variables** such as  $T$ ,  $G$ , or  $M^s$
- Important distinction: **movement along** a curve vs. **shift of** a curve
- Factors that change demand at a given interest rate shift the IS curve. Example: government spending
- An increase in money supply or reduction in reserve requirements shifts the LM curve downwards

## Fiscal Policy

State influence on the economy through fiscal and monetary policy:

- **Fiscal policy** instruments:
  - > Tax reduction/increase: e.g., income tax, VAT
  - > Higher/lower transfer payments: e.g., unemployment benefits
  - > Higher/lower government spending: e.g., infrastructure, education, military

Budget of the state:  $G$  and  $T$  (budget deficit if  $G > T$ )

- **Expansionary** fiscal policy:  $G$  up and/or  $T$  down
- **Contractionary** fiscal policy:  $G$  down and/or  $T$  up
- Changes in  $G$  and  $T$  affect the **IS curve**, not the LM curve

## Expansionary Fiscal Policy (Textbook: Increase in $G$ )

Example: The state increases  $G$  (underutilized capacity), e.g., building new roads.

- Transmission mechanism (interest rate control):
  - >  $G$  up  $\rightarrow$  demand for goods  $Z$  up  $\rightarrow$  production up  $\rightarrow$  income  $Y$  up  $\rightarrow$  **multiplier effect**:  $C$  and  $I$  rise, so  $Y$  continues to rise  $\rightarrow$  demand for money ( $PY * L(i)$ ) up  $\rightarrow$  money supply ( $M_S$ ) up  $\rightarrow$   $i$  remains unchanged
- Expansionary fiscal policy shifts the IS curve to the **right**
- GDP ( $Y$ ) increases by more than the change in  $G$  (multiplier effect)
- The interest rate  $i$  remains unchanged with interest rate control by the central bank

With money supply control, the interest rate  $i$  rises because higher  $Y$  increases money demand while  $M$  is fixed. This **crowds out** private investment:

$$I = I\left(\underset{(+)}{Y}, \underset{(-)}{i}\right) \quad \text{Investment depends negatively on } i$$

- The positive effect on GDP is **smaller** with money supply control than with interest rate control due to crowding out



## Expansionary Fiscal Policy in the IS-LM Model (Graph)

- Graphical representation: Interest rate control vs. money supply control

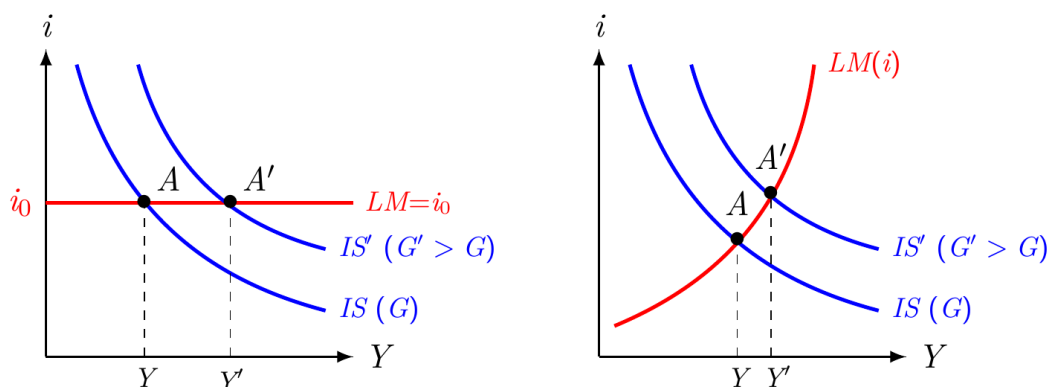


Figure: Expansionary Fiscal Policy - Interest rate control (left: full multiplier) vs. money supply control (right: partial crowding out)

### Practical Example: The Corona Crisis (Lecture)

The corona crisis provides a good example of expansionary fiscal policy. Both government consumption and the SNB balance sheet expanded significantly in 2020.

## Expansionary Fiscal Policy in the IS-LM Model (Practical Example)

- The corona crisis provides a good example of expansionary fiscal policy:

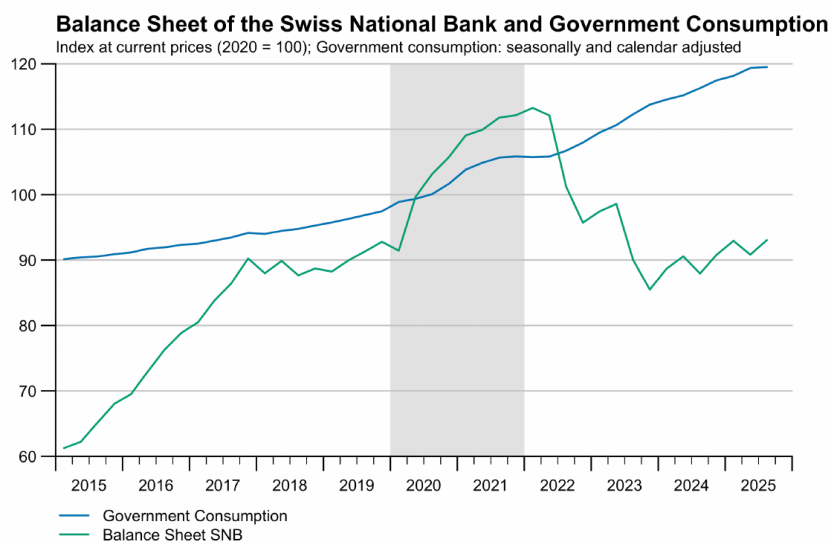


Figure: Swiss National Bank Balance Sheet and Government Consumption (2015-2025, Index 2020=100)

## Contractionary Fiscal Policy (Textbook: Increase in T)

Example: The state raises taxes T.

- Transmission mechanism (interest rate control):

> T up  $\rightarrow$   $Y_D$  down  $\rightarrow$  C down  $\rightarrow$  demand for goods down  $\rightarrow$  Production down  $\rightarrow$  Y down  $\rightarrow$  multiplier effect: C and I fall further  $\rightarrow$  Y falls further  $\rightarrow$  Demand for money ( $PY * L(i)$ ) down  $\rightarrow$  Money supply ( $M_S$ ) down  $\rightarrow$  i unchanged

- Transmission mechanism (money supply control):

> Same initial chain, but: demand for money falls  $\rightarrow$  with  $M_S$  constant  $\rightarrow$  i falls  $\rightarrow$  I up (partially offsetting)

- The decline in GDP due to contractionary fiscal policy is **smaller** if the central bank lets the interest rate fall

### Three Cases of Central Bank Response to Tax Increase (Lecture)

The effect of a tax increase depends critically on how the central bank acts:

**Case 1 - Constant interest rate (i):** Central bank keeps i fixed. A tax increase shifts IS left. To keep i constant, the central bank reduces the money supply. Output falls from  $Y_1$  to  $Y_2$ .

## Contractionary fiscal policy in the IS-LM Model (Graph I)

- Effect of a tax increase depends on how the central bank acts.
- Case 1 — Interest rate control: Central bank keeps interest rate constant

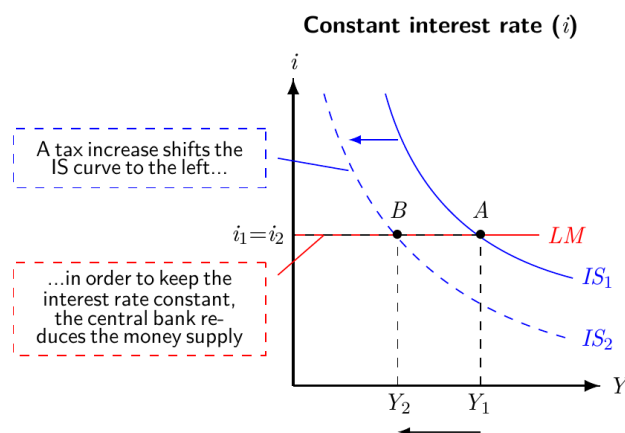


Figure: Contractionary Fiscal Policy, Case 1 - Central bank keeps interest rate constant  $\rightarrow$  full IS multiplier effect

**Case 2 - Constant money supply ( $M^s$ ):** Central bank keeps  $M^s$  fixed. IS shifts left, but since money demand falls with  $Y$ , the interest rate decreases. Output falls less than in Case 1.

## Contractionary fiscal policy in the IS-LM Model (Graph II)

- Case 2 — Money supply control: Central bank keeps money supply constant

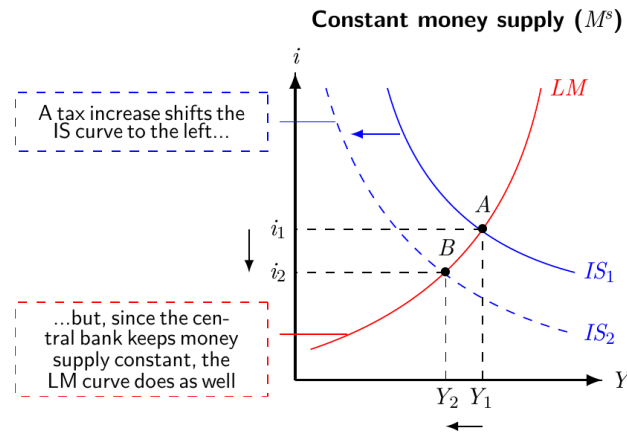


Figure: Contractionary Fiscal Policy, Case 2 - Constant money supply  $\rightarrow$  interest rate falls, partially offsetting output decline

**Case 3 - Constant income (Y):** Central bank wants to keep  $Y$  constant. It lowers the interest rate enough so that higher investment offsets lower consumption.  $Y$  stays at  $Y_1$ , but the composition of GDP changes (less  $C$ , more  $I$ ).

## Contractionary Fiscal Policy in the IS-LM Model (Graph III)

- Case 3 — Income control: Central bank keeps  $Y$  constant

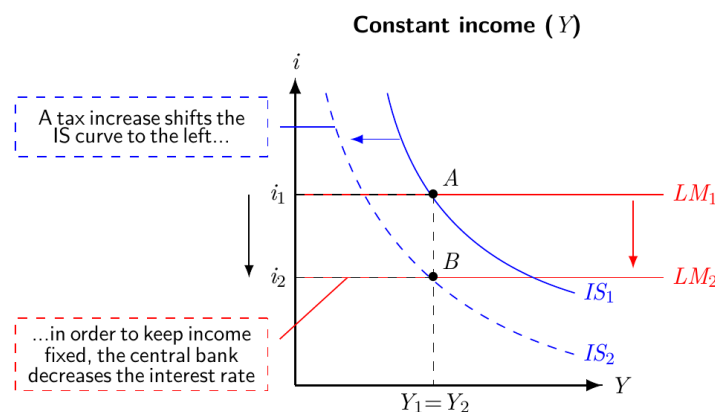


Figure: Contractionary Fiscal Policy, Case 3 - Central bank keeps income constant by lowering interest rate

### Fiscal Policy Discussion (Lecture)

Effects of fiscal policy also depend on the monetary policy the central bank is pursuing. With underutilized capacity,

expansionary fiscal policy can raise  $Y$ . However:

- **Crowding out** of private investment (with money supply control)
- **Value-for-money**: Efficiency of government spending
- **Time lag**: Fiscal policy effects are time-delayed
- **Leakages**: Multiplier reduced by savings and imports
- **Ricardian equivalence**: People expect future tax increases to pay for current spending, so they save more now, reducing the multiplier
- In the medium run, state budget restrictions limit the scope of expansionary fiscal policy

## Monetary Policy

Monetary policy: the central bank's use of interest rates and money supply to influence the economy.

- **Expansionary monetary policy:** A reduction in the interest rate or an increase in the money supply shifts the LM curve **downwards**
- **Contractionary monetary policy:** An increase in the interest rate or a reduction in the money supply shifts the LM curve **upwards**
- Monetary policy does **not** shift the IS curve

### Expansionary Monetary Policy in the IS-LM Model

Example: In March 2020, the US Federal Reserve lowered interest rates.

- Transmission mechanism:
  - >  $M^s$  up  $\rightarrow i$  down  $\rightarrow I(i)$  up  $\rightarrow$  demand for goods up  $\rightarrow Y$  up (production and income)  $\rightarrow$  multiplier effect:  $Y$  up further  $\rightarrow$  demand for money up  $\rightarrow$  Money supply ( $M_S$ ) up
- The central bank adjusts money supply to meet the increased money demand

### Expansionary monetary policy in the IS-LM model

- Example: In March 2020, the US Federal Reserve lowered interest rates
- Transmission mechanism:
  - $M^S \uparrow \rightarrow i \downarrow \rightarrow I(i) \uparrow \rightarrow$  demand for goods  $\uparrow \rightarrow Y$  (production and income)  $\uparrow \rightarrow$  [...multiplier effect...]  $Y \uparrow \rightarrow$  Demand for money ( $PY * L(i)$ )  $\uparrow \rightarrow$  Money supply ( $M_S$ )  $\uparrow$   
(Central bank adjusts the money supply to meet the increased money demand).

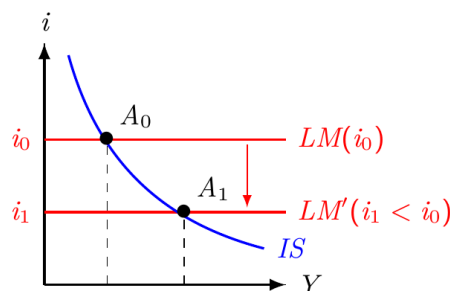


Figure: Expansionary Monetary Policy - LM shifts down from  $LM(i_0)$  to  $LM'(i_1 < i_0)$ , output increases from  $Y_0$  to  $Y_1$

### Monetary Policy Discussion (Lecture)

- Monetary policy has always been controversial
- Key debates:
  - > Does loose monetary policy generate inflation? ( $\rightarrow$  Lecture 6)
  - > **Liquidity trap:** What happens when interest rates are already at zero?
  - > **Credibility** of the central bank (e.g., Greenspan put, moral hazard)
  - > Effects on the **exchange rate** ( $\rightarrow$  Lectures 8 and 9)

- SNB: room for maneuver on interest rates exhausted after the financial crisis, hence accumulation of foreign exchange reserves (purchase of foreign assets) to reduce appreciation pressure

## The Liquidity Trap

Very low interest rates since the financial crisis have raised concerns about the liquidity trap. When interest rates are at or near zero:

- Additional increases in money supply have no effect on the interest rate (people are willing to hold any amount of money at  $i = 0$ )
- Monetary policy becomes **ineffective** - the LM curve becomes flat
- Fiscal policy must do the heavy lifting

## Limits of monetary policy — during the Corona crisis

- Very low interest rates since the financial crisis:

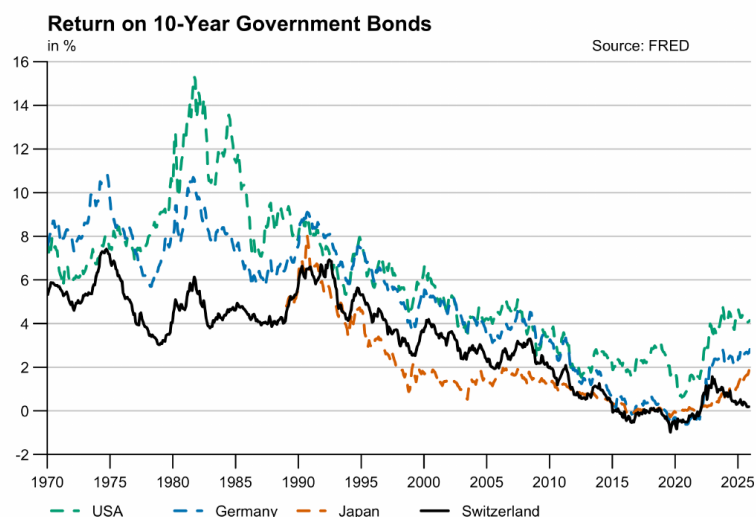


Figure: Return on 10-Year Government Bonds (1970-2025) for USA, Germany, Japan, and Switzerland (Source: FRED)

## The Liquidity Trap (Graph)

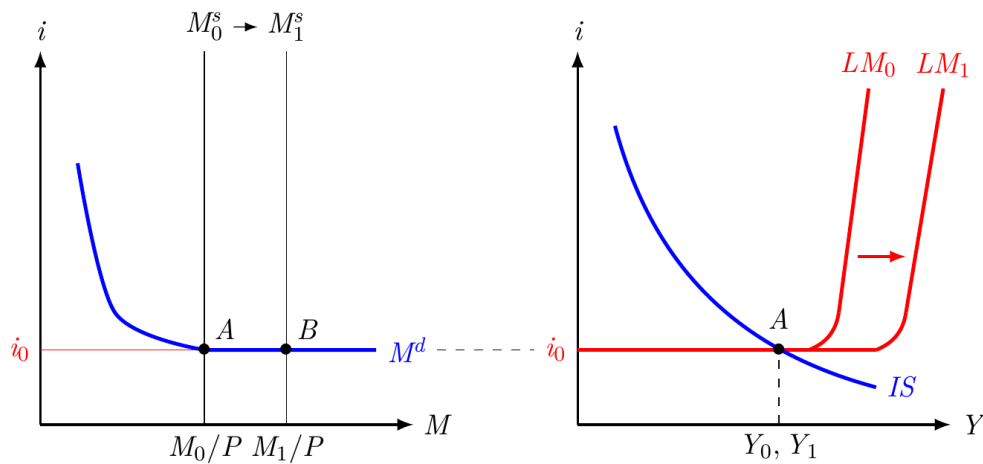


Figure: The Liquidity Trap - At  $i_0$  near zero, increasing money supply from  $M_0$  to  $M_1$  has no effect on output ( $Y_0 = Y_1$ )

## 5-4 Using a Policy Mix

In practice, fiscal and monetary policies are often used together. The combination is called the monetary-fiscal policy mix, or simply the policy mix.

### Same Direction Policy Mix

- When the economy is in recession and output is too low, both policies can be used to increase output
- Expansionary fiscal policy (e.g., decrease in  $T$ ) shifts  $IS$  to the right
- Expansionary monetary policy (decrease in  $i$ ) shifts  $LM$  down
- Both contribute to higher output. Higher income and lower taxes raise consumption; higher output and lower interest rate raise investment

### Opposite Direction Policy Mix

Sometimes the right mix uses the two policies in opposite directions. Example: fiscal consolidation combined with monetary expansion.

- Government wants to reduce the deficit ( $T$  up or  $G$  down)  $\rightarrow$   $IS$  shifts left to  $IS'$
- By itself, this would cause a recession (output falls from  $Y$  to  $Y'$ )
- Central bank reduces the interest rate  $\rightarrow$   $LM$  shifts down to  $LM'$
- The combination achieves: deficit reduction + unchanged (or higher) output
- Investment unambiguously increases (lower interest rate), while consumption effects depend on method (tax increase vs. spending cuts)

### Why Use a Policy Mix?

(1) Running large deficits is dangerous - better to share the burden with monetary policy. (2) Fiscal and monetary policies have

different effects on output composition (C vs I). (3) Neither policy works perfectly alone - a decrease in taxes may fail to increase consumption; a decrease in the interest rate may fail to increase investment. Using both hedges against failure.

## 5-5 How Does the IS-LM Model Fit the Facts?

The IS-LM model so far has ignored dynamics: adjustment takes time, not instantaneous.

- Consumers take time to adjust consumption after changes in disposable income
- Firms take time to adjust investment after changes in sales or interest rates
- Firms take time to adjust production after changes in sales

### Empirical Evidence (Lecture: Chapter 5.5)

Econometric studies (Christiano, Eichenbaum, Evans, 1996) using U.S. data from 1960-1990 show that a 1% increase in the federal funds rate leads to:

- A decline in retail sales, with the largest decrease (-0.9%) after 5 quarters
- A decline in output, with the largest decrease (-0.7%) after 8 quarters
- A decline in employment, slow and steady, reaching -0.5% after 8 quarters
- An increase in the unemployment rate
- Nearly no change in the price level for the first 6 quarters (validating the IS-LM assumption of fixed prices in the short run)

Key takeaway: Monetary policy works, but with long lags. It takes nearly two years for the full effect on output.

#### Empirical Validation

Various empirical studies support the core statements of the IS-LM model and also show the dynamics (time lag). However, large isolated shocks and policy changes are rare, making empirical studies difficult (-> chapter 5.5).



## CHAPTER 6: FINANCIAL MARKETS II - THE EXTENDED IS-LM MODEL

So far we assumed only two financial assets (money and bonds) and one interest rate. Reality is more complex: there are many interest rates, many financial institutions, and the financial system plays a major role in the economy (7% of U.S. GDP). The 2008 crisis showed this assumption was too simplistic.

### 6-1 Nominal versus Real Interest Rates

#### Definitions

- **Nominal interest rate ( $i$ )**: Interest rate in terms of dollars (or national currency). What you see in newspapers.
- **Real interest rate ( $r$ )**: Interest rate in terms of goods. What matters for consumption and investment decisions.

#### Derivation of the Real Interest Rate

The exact relation between nominal and real interest rates:

$$(6.1) \quad (1 + r_t) = (1 + i_t) \cdot \frac{P_t}{P_{t+1}^e}$$

Defining expected inflation:

$$(6.2) \quad \pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t}$$

This gives us:

$$(6.3) \quad (1 + r_t) = \frac{1 + i_t}{1 + \pi_{t+1}^e}$$

When inflation is not too large (< 20%), we can use the approximation:

$$(6.4) \quad r_t \approx i_t - \pi_{t+1}^e$$

#### Real Interest Rate

The real interest rate is approximately equal to the nominal interest rate minus expected inflation. When expected inflation = 0, nominal = real rate. Because expected inflation is typically positive, the real rate is typically lower than the nominal rate. The real rate is what matters for spending decisions.

#### The Zero Lower Bound and Deflation

- The nominal interest rate cannot be negative (or only very slightly): **zero lower bound (ZLB)**
- The real interest rate cannot be lower than minus expected inflation:  $r \geq -\pi^e$
- If expected inflation is positive (e.g., 2%), the real rate can go as low as -2%
- If expected inflation turns **negative** (deflation), the lower bound on the real rate is **positive**, which may not be

low enough to stimulate the economy -> serious concern during the 2008 crisis

## 6-2 Risk and Risk Premia

Not all bonds are the same. They differ in maturity and risk of default.

- **Risk premium (x)**: The additional interest rate charged on risky bonds to compensate for default risk

The risk premium depends on two factors:

- (1) **Probability of default (p)**: Higher p -> higher risk premium

$$(1 + i) = (1 - p)(1 + i + x) + (p)(0)$$

$$x = \frac{(1 + i) \cdot p}{1 - p} \approx p \quad (\text{for small } i, p)$$

- (2) **Risk aversion**: Even with fair expected returns, risk-averse investors demand extra compensation
- During the 2008 crisis: AAA bond rates rose to 8%, BBB rates to 10%, while government bond rates fell (flight to safety)

## 6-3 The Role of Financial Intermediaries

Most borrowing happens through financial intermediaries (banks, non-banks, hedge funds), not through direct bond issuance.

### Leverage and Its Consequences

- **Capital ratio** = capital / assets (e.g., 20/100 = 20%)
- **Leverage ratio** = assets / capital (e.g., 100/20 = 5)
- Higher leverage -> higher expected profit per unit of capital, but higher risk of **insolvency**
- When asset values decline, highly leveraged banks must either raise capital or reduce lending -> **credit crunch**

### Liquidity Risk and Bank Runs

- Banks' assets (loans) are **illiquid** - hard to sell quickly
- Banks' liabilities (deposits) are **liquid** - can be withdrawn at short notice
- If investors doubt asset values -> they withdraw funds -> bank must sell at **fire sale prices** -> further losses -> potential insolvency
- This can be self-fulfilling: even a solvent bank can fail if enough depositors panic
- Solutions: **federal deposit insurance** (FDIC), **liquidity provision** by central bank, **narrow banking** proposals

## 6-4 Extending the IS-LM

We now extend the IS-LM model to account for the difference between nominal and real rates, and between the policy rate and the borrowing rate:

$$\text{IS relation: } Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

$$\text{LM relation: } i = \bar{i}$$

Simplifying (central bank chooses the real policy rate  $r$ ):

$$(6.5) \quad \text{IS: } Y = C(Y - T) + I(Y, r + x) + G$$

$$(6.6) \quad \text{LM: } r = \bar{r}$$

- The **policy rate ( $r$ )** is set by the central bank (enters the LM relation)
- The **borrowing rate ( $r + x$ )** is what consumers and firms actually face (enters the IS relation)
- Even if the central bank keeps  $r$  constant, an increase in the risk premium  $x$  raises the borrowing rate and shifts the IS curve to the left

### Financial Shocks and Policy Responses

- An increase in risk premium  $x \rightarrow$  IS shifts left  $\rightarrow$  output falls (financial crisis becomes macroeconomic crisis)
- Monetary policy response: decrease  $r$  to offset the increase in  $x$
- But: the ZLB limits how far  $r$  can fall. The lowest real policy rate =  $-\pi^e$
- If inflation is low, the central bank may not be able to fully offset a large financial shock

## 6-5 From a Housing Problem to a Financial Crisis

The 2008 financial crisis illustrates how a housing price decline was amplified by the financial system.

### Key Amplification Mechanisms

- **High leverage:** Banks underestimated risk; compensation systems incentivized risk-taking; SIVs were used to circumvent capital requirements
- **Securitization:** Mortgages bundled into MBS and CDOs. Reduced incentives for careful lending; rating agencies underestimated risk; assets became **toxic** - extremely hard to value
- **Wholesale funding:** Banks relied on short-term borrowing from other investors (not just deposits). When confidence collapsed, this funding dried up.
- Result: High leverage + illiquid assets + liquid liabilities  $\rightarrow$  runs on financial institutions, fire sales, credit crunch

### Policy Responses to the Crisis

- **Financial policies:** Deposit insurance increased (\$250K), Fed provided liquidity facilities, TARP (\$700B) to recapitalize banks
- **Monetary policy:** Fed cut rates to zero by Dec 2008, then turned to unconventional policy (QE) - buying long-term assets to reduce borrowing rates
- **Fiscal policy:** American Recovery and Reinvestment Act (\$780B in 2009) - tax cuts and spending increases
- The initial shock was so large that even combined financial, monetary, and fiscal policies were not enough to avoid a major recession (U.S. GDP fell 3.5% in 2009)

# BUSINESS CYCLES

## GDP Fluctuations

The goods market (GDP) is subject to fluctuations over time. Example: Swiss GDP since 2000 shows a long-term trend of approximately 1.9% growth per year, with short-term fluctuations around this trend.

### Business cycles

- ▶ The goods market (the GDP) is subject to fluctuations over time.
- ▶ Example: Switzerland's GDP since 2000

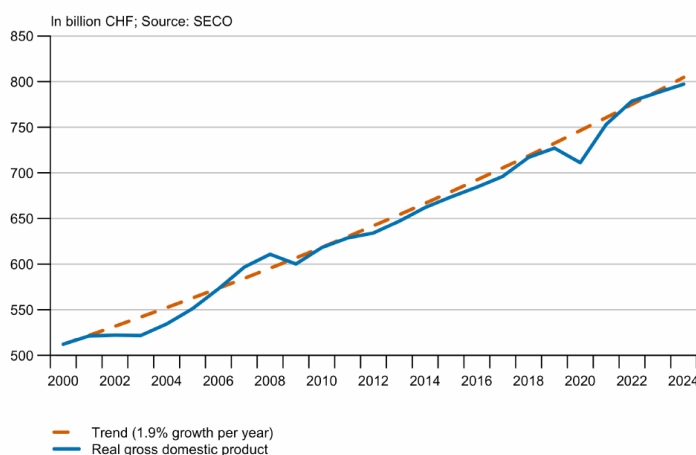


Figure: Switzerland's GDP since 2000 - Real GDP vs. 1.9% trend line (Source: SECO)

## Explanatory Approaches

$$\text{GDP: } Y = C + I + G + NX$$

- Fluctuations in GDP can have many causes - many factors determine C, I, G, NX
- General framework: **Impulse/shock** -> transmission -> interaction -> economic fluctuation
- Possible stimuli: fiscal & monetary policy measures, technological leaps, demand shocks, financial crises (Minsky hypothesis)
- The IS-LM model helps with:
  - > Analysis of business cycles
  - > Forecast of short-run GDP development (and its difficulty!)

## Sector Correlations with GDP Growth (Swiss Data)

Not all industries are affected by the business cycle to the same extent. A positive correlation indicates the sector is pro-cyclical:

| Sector                                  | Correlation |
|-----------------------------------------|-------------|
| Manufacturing (Noga 10-33)              | 0.70        |
| Business services (Noga 68-75, 77-82)   | 0.57        |
| Hospitality (Noga 55-56)                | 0.42        |
| Retail trade (Noga 47)                  | 0.39        |
| Health and social services (Noga 86-88) | 0.15        |
| Education (Noga 85)                     | 0.05        |
| Construction (Noga 41-43)               | 0.05        |
| Water, sewage, waste (Noga 36-39)       | -0.42       |

Real, seasonally, calendar and sports-event adjusted, 1990-2024 (excl. Covid years 2020 & 2021). Data: SECO.

## Extensions and Limitations of the IS-LM Model

The IS-LM model can be extended in many ways:

- **Nominal vs real interest rates and risk premiums** (Chapter 6)
- **Consumption:**  $C = C(Y_D)$  may also depend on expectations (consumer confidence) or assets (IS-LM model ignores stocks)
- **Investments:** May depend on expectations (-> PMI, investment trap)
- **Open economy:** Exports and imports, exchange rates

Important three steps: Understanding the model, applying the model, questioning the model.

## Limitations and Open Questions

- How do the short- and medium-run effects of expansionary or contractionary fiscal policy differ?
- Should the state better reduce taxes (T down) or increase spending (G up)?
- What else can the central bank do when the interest rate is already at zero?
- Are the incentives for key decision-makers in fiscal and monetary policy identical to those of the population? (-> political economy)
- In the model, G and thus Y can be increased at will - what is the limit?
- In the **medium run**, resource constraints are important; prices are not stable and inflation can occur due to expansionary policy
- -> In lectures 5 and 6 we will examine the labor market and inflation

## SUMMARY OF KEY EQUATIONS AND TERMS

### Key Equations - Chapter 5

$$(5.1) \quad I = I(Y, i) \quad \text{Investment relation}$$

(+)      (-)

$$(5.2) \quad Y = C(Y - T) + I(Y, i) + G \quad \text{IS relation}$$

$$(5.3) \quad \frac{M}{P} = Y \cdot L(i) \quad \text{LM relation (real terms)}$$

$$i = \bar{i} \quad \text{LM curve (interest rate control)}$$

### Key Equations - Chapter 6

$$(6.1) \quad (1 + r_t) = (1 + i_t) \cdot \frac{P_t}{P_{t+1}^e} \quad \text{Exact real interest rate}$$

$$(6.4) \quad r_t \approx i_t - \pi_{t+1}^e \quad \text{Approximate real interest rate}$$

$$x \approx p \quad \text{Risk premium} \approx \text{probability of default}$$

$$(6.5) \quad Y = C(Y - T) + I(Y, r + x) + G \quad \text{Extended IS relation}$$

$$(6.6) \quad r = \bar{r} \quad \text{Extended LM relation}$$

### Policy Effects Summary

| Policy                    | IS curve | LM curve | Output Y |
|---------------------------|----------|----------|----------|
| G up (fiscal expansion)   | Right    | No shift | Up       |
| T up (fiscal contraction) | Left     | No shift | Down     |
| i down (monetary exp.)    | No shift | Down     | Up       |
| i up (monetary contr.)    | No shift | Up       | Down     |
| x up (financial shock)    | Left     | No shift | Down     |

## Key Terms

- **IS curve:** Downward-sloping; all  $(i, Y)$  pairs where goods market is in equilibrium
- **LM curve (modern):** Horizontal line at  $i$ -bar; financial market equilibrium under interest rate control
- **LM curve (traditional):** Upward-sloping; financial market equilibrium under money supply control
- **Fiscal contraction/consolidation:** Increase in  $T$  or decrease in  $G$ ; reduces budget deficit
- **Fiscal expansion:** Decrease in  $T$  or increase in  $G$ ; increases budget deficit
- **Monetary expansion:** Decrease in  $i$  (increase in  $M$ ); shifts LM down
- **Monetary contraction/tightening:** Increase in  $i$  (decrease in  $M$ ); shifts LM up
- **Policy mix:** Combination of fiscal and monetary policy
- **Nominal interest rate ( $i$ ):** Interest rate in terms of currency (dollars)
- **Real interest rate ( $r$ ):** Interest rate in terms of goods:  $r = i - \pi^e$
- **Risk premium ( $x$ ):** Extra interest charged for default risk and risk aversion
- **Policy rate ( $r$ -bar):** Real interest rate chosen by the central bank
- **Borrowing rate ( $r + x$ ):** Rate actually faced by consumers and firms
- **Capital ratio:** Capital / Assets (e.g., 20%)
- **Leverage ratio:** Assets / Capital (e.g., 5x); inverse of capital ratio
- **Insolvency:** When assets < liabilities; bank goes bankrupt
- **Fire sale prices:** Prices far below true value; forced selling
- **Bank run:** Depositors panic and withdraw; can cause even solvent banks to fail
- **Securitization:** Bundling assets (e.g., mortgages) into tradable securities (MBS, CDOs)
- **Subprime mortgages:** Risky mortgages to borrowers with poor credit
- **Underwater mortgage:** Mortgage value > house value
- **Wholesale funding:** Banks borrowing from investors (not deposits); liquid liabilities
- **TARP:** Troubled Asset Relief Program (\$700B); used to recapitalize U.S. banks
- **Unconventional monetary policy:** QE, buying long-term assets when at ZLB
- **Crowding out:** Government spending raises  $i$ , which reduces private investment
- **Ricardian equivalence:** People expect future taxes to offset current deficits, reducing multiplier
- **Liquidity trap:** At  $i = 0$ , more money has no effect; monetary policy is ineffective
- **Confidence band:** Range within which the true effect lies with 60% probability